

Enhancing Telecommunications Network Availability through Infrastructure Protection with High-Spatial-Resolution Sensing

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Abstract *A framework for increasing optical fiber network availability by integrating high-spatial-resolution distributed fiber sensing in data transmission systems is presented. Coexistence with parallel data channels on a separate wavelength, mitigation of nonlinear signal interaction impacts, applications and challenges are investigated. ©2026 The Author(s)*

Introduction

Optical fiber networks form the backbone of today's internet, providing high-capacity, low-latency, and long-range transmission required for cloud computing, interconnecting data centers, mobile backhaul architectures, and artificial intelligence tasks. Therefore, high availability of this critical infrastructure is crucial.

In addition to transmitting data, optical fibers can also be used as sensors. Environmental effects like vibrations, acoustic signals, strain and temperature changes in close proximity to a fiber impact its properties in terms of the refractive index and length. These changes can be detected by launching probe light into the fiber and evaluating the changes of the received electrical field with respect to amplitude, phase and polarization. In particular, the optical phase is the most sensitive parameter, and it can be evaluated using feed-forward type sensing, where transmitter and receiver are located at the two ends of a fiber link, and with single-sided back-scattering type methods, such as coherent optical time-domain reflectometry (C-OTDR).

The integration of fiber sensing in a data carrying fiber link on a separate wavelength can support early identification of events in the vicinity of the fiber that pose a risk to the link integrity, such as a fiber cut. The goal is to use fiber sensing technologies to reduce link outages and maintenance costs, thereby increasing network availability. High-spatial-resolution sensing methods have the potential for more accurately classifying the event type and its location, helping to distinguish critical from non-critical events. This supports network owners and operators in taking corresponding actions.

Applications of fiber sensing in communications networks

In terrestrial networks, early detection of earthworks close to an underground fiber cable and failures of mechanical cable mounting parts, e.g. as used in aerial cables, is an important step towards higher network reliability. Such mechanical

parts can fail due to wear accelerated by extreme weather conditions. Furthermore, the detection of unauthorized access to and sabotage of the fiber infrastructure, which has happened more frequently in the recent past and has led to massive outages, has become more relevant.

In subsea links, detection of trawling close to a fiber cable and anchor dragging by malicious actors is of concern [1]. In addition, fiber sensing in subsea cables can be used to detect "dark vessels" with a deactivated automatic identification system [2].

This highlights that sensing on already deployed telecom fiber cables not only helps to protect the fiber infrastructure itself, but also protecting other critical infrastructure deployed in close proximity to the fiber. Such infrastructures are roads, railways, power lines, water, gas and oil pipelines. Furthermore, sensing on subsea fibers is used for seismic activity monitoring and subsea wildlife research [3].

Sensing integration and process chain for increasing network availability

A schematic for integrating sensing and the processing chain required to improve network availability is shown in Fig. 1. In addition to the optical fiber itself, communication networks consist of transceivers and network elements such as optical amplifiers and reconfigurable optical add-drop multiplexers, and these components can deliver telemetry data. In addition, monitoring the state of polarization (SoP) in coherent receivers by extracting the Stokes parameters directly from the digital signal processor (DSP) is gaining more traction, since this allows for feed-forward sensing without significant costs of installation. However, SoP sensing is limited by its low sensitivity to external perturbations compared to phase sensing.

Coarsely resolving the location of an event is possible in feed-forward type sensing by using a bidirectional configuration and comparing the time instances at which an event is visible at each end of the link [4]. However, the spatial resolution capabilities of this concept depend on the frequency components of the event and can be in

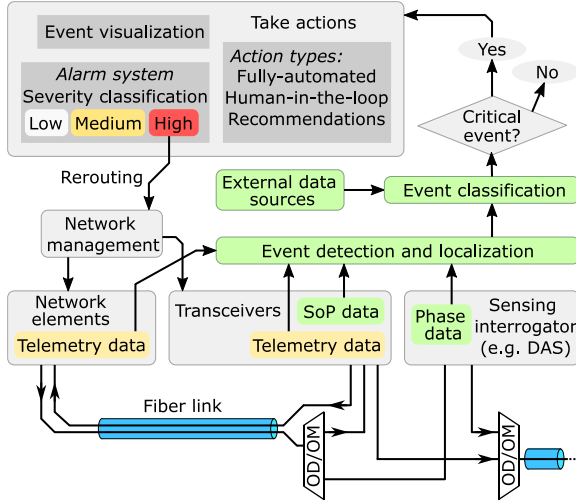


Fig. 1: Multi-modal sensing in a fiber link and process to increase network availability (OD/OM: optical demultiplexer/multiplexer)

the order of several kilometers. Complementing a critical link with distributed sensing, such as distributed acoustic sensing (DAS), with high spatial resolution in the meter range, will significantly improve localization capabilities, and the extracted phase information has a significantly higher sensitivity compared to polarization-based sensing.

The phase data from a sensing interrogator as well as the SoP and telemetry data from sensing-enabled transceivers can be combined to detect and localize anomalies in a fiber link. In a second stage, trained machine learning (ML) models can be used to classify the event type, and based on this, a decision on the criticality of the event for the integrity of the fiber infrastructure can be made. For instance, vibrations induced by a car driving on a street next to a deployed fiber are not critical, but vibrations from earthworks and malicious attacks, which may involve direct contact with the cable, can pose a risk to the fiber infrastructure. In the latter cases, corresponding actions are taken such as notifying the operator of the nature of the event with suggested actions. This notification may comprise suggestions for repair actions, for instance in case of defects in cable mounting parts. Actions can also be automatically triggered, e.g. rerouting the data traffic in case of immediate risk of a link outage.

High-spatial-resolution sensing

Increasing the spatial resolution of back-scattering-based distributed sensing provides more information with respect to regional spread and pattern of an event. In other words, the fiber is divided into shorter segments, where each segment can be considered as an individual sensor. Correspondingly, a higher spatial resolution leads to a higher number of sensors along the fiber link.

Results of an exemplary measurement with a high spatial resolution using a coherent

correlation OTDR (CC-OTDR) in the form of phase differences of consecutive fiber segments and the corresponding setup are shown in Fig. 2 [5]. Herein, a hammer impact on the ground near a deployed underground fiber cable generates a pressure wave. This wave is visible in the fiber region between 327.5 m and 345.3 m. This region is divided into six segments, providing an average spatial resolution of 3 meters. Instances at which the pressure waves arrive at each segment can be derived from the phase difference traces of the individual segments. In segment 4, between 336.3 m and 339.6 m, the variation from the impact is visible earliest, indicating that this section is the closest to the impact. Correspondingly, the phase variation magnitude is the highest within this section, and it decays the further away the sensor segments get from this section. Based on the start time of the variation in the segments, the propagation speed of the pressure wave can be extracted. This kind of additional spatial information helps to train ML models to more precisely classify events.

In conventional DAS using pulses as probe signals, a high spatial resolution is achieved by using short probe pulse durations, which significantly limit the sensing range. By modulating the probe pulse with e.g. a code sequence or by chirping the pulse, a long probe signal duration in the order of tens of microseconds can be used, enabling a long sensing range. At the same time, a high spatial resolution is achieved when using pulse compression by correlating the returned back-scattered signal with the transmitted probe signal.

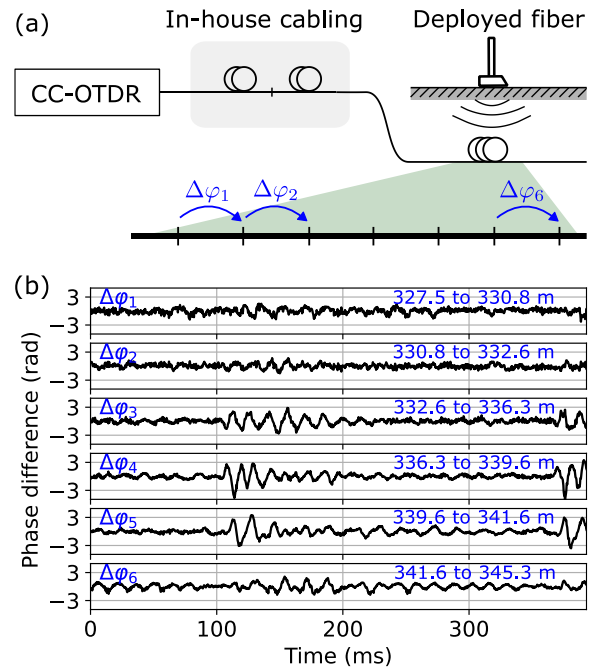


Fig. 2: Ground impact induced pressure wave affecting a deployed underground cable: (a) Testbed setup and (b) measured phase differences of consecutive fiber segments

Integration in communications networks

The integration of sensing in a dense wavelength division multiplexing (DWDM) fiber link can be performed by transmitting the probe signal on a separate wavelength. Therefore, a free wavelength channel of the ITU-T grid in the C-band or L-band can be used (in-band sensing) or a wavelength outside of this region in the S-, upper L- or U-band can be used (out-of band sensing).

Recently, the ITU-T G.681 recommendation has been approved, which specifies the system architecture and optical parameters to ensure that integration of sensing has a negligible effect on the transmission performance [6]. In amplified links, the optical signal-to-noise ratio required (rOSNR) to achieve quasi bit-error rate free transmission after forward error correction (FEC) is used as a metric. When integrating sensing into a DWDM link, the degradation of the rOSNR should not be larger than 0.1 dB. This integration can be done in two ways, co-directionally and counter-directionally. Here, the data signal and the sensing probe signal propagate in the fiber in the same or opposite direction, respectively. Several experiments have been conducted to evaluate the penalty on the rOSNR for both configurations [7,8]. In these tests, the sensing probe was transmitted at 193.4 THz and the data channel was located at 50 and 100 GHz offsets. At such small wavelength spacings, the effect of cross-phase modulation (XPM) from the bursty probe signal on the data signals is dominant.

In case of co-directional propagation, the effect of XPM accumulates in a relatively small section of the data signal affected by the bursty probe signal propagating in parallel due to small walk-off, which leads to a phase shift in the data signal. At the edges of the probe signal with a pulse-like power envelope, this leads to fast phase transients in the data signal, and it is difficult for the receiver DSP to compensate for this effect. This leads to a measured rOSNR penalty of 0.4 dB for DP-QPSK (100G) for typical power levels, whereas for DP-16QAM (200G) the DSP cannot correct this effect, leading to uncorrected blocks after FEC. By using waveform shaping of the probe signal to reduce the phase transient speed, the impact on the rOSNR can be significantly reduced, as illustrated in Tab. 1.

In contrast, for counter-directional integration, each section of the data signal sees the probe pulse only for a short time, such that all sections are affected in a similar way, which avoids the fast phase transients. It leads to no measurable impact on the rOSNR when the CC-OTDR is switched on, see Tab. 1(b).

This shows that with further optimizations and proper signal waveform design, co-directional

and counter-directional integration of sensing in a fiber link is possible with minimal impact on the transmission quality. At larger frequency spacings in the order of 13 THz, Raman induced power transfers between the sensing signal and data signals need to be considered.

Tab. 1: Required OSNR penalty compared to the case with CC-OTDR being switched off

(a) Co-directional integration at 100 GHz spacing

Mode	rOSNR penalty
DP-16QAM CC-OTDR on	Uncorrected blocks
DP-16QAM CC-OTDR on (shaping)	0.4 dB
DP-QPSK CC-OTDR on	0.4 dB
DP-QPSK CC-OTDR on (shaping)	0 dB

(b) Counter-directional integration at 50 GHz spacing

Mode	rOSNR penalty
DP-16QAM CC-OTDR on	0 dB

Challenges

The following items need to be improved to effectively use fiber sensing for increasing the availability of telecom networks. Firstly, it has been shown that with sensing data produced by interrogators, events like vibrations, acoustic signals, temperature and strain can be detected. However, the interpretation of this data for classifying and identifying events that pose a risk to the integrity of a fiber infrastructure is a challenge. For instance, vibrations from weather effects like wind and rain are visible. Furthermore, cars driving on a road that is routed alongside a deployed fiber cable also produce vibrations. However, these events are not critical, yet they are frequently visible in the measured data. In contrast, critical events like vibrations from earthworks close to a deployed fiber cable need to be distinguished from other non-critical vibration events. For effective identification, pattern recognition methods e.g. based on trained ML-models are crucial. A further challenge is the high amount of raw measurement data that is produced, which requires effective signal processing and data reduction algorithms without losing relevant data. Finally, the scaling of sensing systems by reducing the cost per monitored fiber is key for their adoption in critical fiber infrastructures.

Conclusions

Techniques for integrating fiber sensing into communications networks and the corresponding process framework for increasing network availability are presented. High-spatial-resolution fiber sensing methods can help to improve event classification, enabling to differentiate critical from non-critical events. Applications, methods for integrating sensing without affecting transmission quality and open challenges are presented.

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